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The Editor-in-Chief
Journal of Visual Communication and Image Representation
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Dear Editor-in-Chief,

We respectfully submit our manuscript, **"Net-Aware Learned Block Mapping for Reversible Data Hiding in Binary Images,"** for consideration as a **Research Article** in the *Journal of Visual Communication and Image Representation.*

Scope Fitness Assessment. The manuscript addresses reversible visual-data representation in binary images through perceptual distortion control, exact restoration, learned content analysis, and auditable payload measurement:

- **Image representation problem:** application data are embedded in binary images with bitwise recovery of the message and original cover.
- **Technical contribution:** globally injective distance-one substitutions, CNN perceptual ranking, and enumerative coding measure useful capacity after serialized metadata.
- **Common protocol:** ABM, PPOCP, Huynh–Nguyen, and Dong use identical covers, messages, payloads, distortion, accounting, and round-trip checks.

Scientific Novelty and Evidence. The central contribution is the systematic criterion $C_{\text{net}} = C_{\text{gross}} - |A_{\text{wire}}|$, which prevents side information from being excluded from capacity claims. Eight-fold leave-one-image-out validation covers 13,808 candidates ($r = 0.989$); held-out placement reduces mean DRD from 0.8054 to 0.7663 (4.85%) and improves every document image. The mapping layer carries 2245 net bits on average on eight document benchmarks and reaches the highest observed net-payload operating point on matched BOSSbase images. Availability, Holm-corrected inference, runtime, memory, and grouped steganalysis are also reported.

Practical and Reproducibility Value. The work provides reversibility proofs, deterministic algorithms, an exact wire format, model weights, tests, per-image results, plotting scripts, and fixed seeds. The reproducibility package is available at github.com/EkodeckStephane/net-aware-block-mapping-rdh.

Mandatory Author Declarations.

- **Originality and exclusivity:** the manuscript is original, unpublished, and not under consideration elsewhere.
- **Author approval:** all authors approved the manuscript, authorship order, contributions, and submission.
- **Interests and funding:** there are no competing interests or external funding.
- **Ethics:** no humans, animals, clinical interventions, or sensitive personal data are involved.

The manuscript will interest JVCIR readers working on image representation, reversible data hiding, perceptual quality, and learned image analysis. Thank you for considering our submission.

Yours sincerely,

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